June 16, 2025

Centers for Medicare & Medicaid Services
Department of Health and Human Services
Attention: CMS-2025-0050-0031
P.O. Box 8013
Baltimore, MD 21244–8013

Regarding CMS-2025-0050-0031, Request for Information (RFI) Health Technology Ecosystem

Submitted electronically at: https://www.regulations.gov/commenton/CMS-2025-0050-0031

To the Director of the Centers for Medicare & Medicaid Services (CMS):

Preamble

eHealth Exchange appreciates the opportunity to respond to Centers for Medicare & Medicaid Services as it seeks to refine its vision for the next stages of a national health data sharing strategy. The HHS/CMS founded eHealth Exchange network (then known as the NHIN) in 2006. Since then, eHealth Exchange, a 501(c)(3) non-profit for the public good, has been working diligently to realize the vision set out by the CMS of providing a secure national gateway for all authorized actors in the healthcare ecosystem, including patients, payers and providers. We have grown to connect the vast majority of the United States, with zero government or private grants. As such, we have unique experience in implementing exchange at scale that we are happy to publicly share for the common good.

eHealth Exchange wants to recognize and commend CMS for sponsoring FHIR Connectathons and supporting industry standards development. eHealth Exchange has been an avid supporter of CMS Connectathons and has brought systems to test at every CMS Connectathon since the first event five years ago. We encourage CMS to continue sponsoring these virtual and in-person events. They are an effective way for CMS to engage with the broader community, encourage innovation, and gauge market readiness for adoption of modern technologies.

We are at an important stage of our national strategy; as a country we have invested tens of billions of dollars in moving health systems out of the paper world into electronic record keeping. We have also invested in establishing broad national connectivity. But fragmentation has occurred since the NHIN was envisioned as the national healthcare "dial tone". Competing standards, conflicting regulations at the state and federal level, and different rates of progress from different participants in the ecosystem have resulted in some actors being unable to connect to national networks, yet.

To help connect some of these siloed groups, over the past three years, eHealth Exchange has invested heavily in improving production exchange for Public Health, Payers, Patients, and Providers. These efforts have yielded some pragmatic successes that can perhaps help inform CMS with specific production examples demonstrating which exchange methods can be adopted at national scale. However, these leading-edge efforts have also, as expected, resulted in identification of a number of impediments to achieving the next generation of national health data exchange. We have prepared this document to share the successful lessons learned as well as the barriers that CMS, and the broader community, can work towards overcoming.

Later in this document, we provide line-by-line responses to many specific questions asked in the CMS RFI. But before we get into that level of granularity, we first wanted to provide some broad-brush comments, and then to highlight some specific production case studies to inform the CMS strategy based on real word experiences.

Broad Themes

1. The CMS should facilitate the creation of an updated national directory of all healthcare and payment actors. A usable directory of actors, and the relationships between those actors, is a necessary prerequisite for the next phase of national data exchange.

   The current CMS mandated FHIR Provider and Plan Directory API end points are not sufficient. Directory end points/APIs have significant differences in FHIR data models, inconsistent support of obtaining updates, different fundamental methods of representing provider to organization relationships, lack of technical support by the publishing organizations, rate limiting resulting in weeks of time needed to obtain directory data, inappropriate authentication barriers, protracted downtime (months in some cases), data accuracy, and more. Private sector-maintained directories are sometimes not granular enough to represent the actual person or system making the request, and they are often incomplete, or out-of-date. This results in each directory connection being non-scalable and an expensive "one off" with unclear data quality.

   If providing a single national directory is not aligned with CMS's direction, then CMS should at least publish an accurate "directory of directories" listing each provider / payer / plan directory API endpoint. Today, developers are forced to use manual means to discover these end points. This manual discovery process is expensive, time consuming, and reduces developer interest due to the resources needed to find each FHIR directory API.

   Additionally, the CMS should mandate that ALL private and public payers, and all provider organizations publish FHIR directory APIs. Notably, some states do not publish their directories using the CMS FHIR directory standards resulting in state-specific one-off solutions to obtain directory information.

   Providing a viable single national directory, or at least a directory of directories, would likely reduce costs to all health data sharing actors by standardizing these important data and reducing manual one-off integration work. A directory, once deployed, would accelerate developer interest in creating applications since they could now easily identify the actors, including their FHIR API end points, in the national healthcare ecosystem with only a trivial amount of effort. Today, identifying these FHIR API end points is an almost insurmountable challenge. The CMS is in a unique position to solve this problem.

   Finally, the CMS should fund work at IHE USA and HL7 standards bodies for the creation of a more constrained FHIR Directory Implementation Guide to eliminate the variations between APIs so that true directory interoperability is achieved and expensive one-off integrations avoided. CMS should also investigate leveraging existing directories such as those curated by the national networks and

frameworks (eHealth Exchange, TEFCA, Carequality) and Council for Affordable Quality Healthcare (CAQH).

2. eHealth Exchange is currently demonstrating that payers can "plug in" to national networks and reap the benefits of the decade of investment we have made in interoperability. Below, we provide a compelling case study illustrating the use of standards-based "Networked FHIR" as a viable and scalable approach for significantly improving care coordination and payment transactions at scale. eHealth Exchange has deployed the first FHIR-based national network electronic Prior Authorization (ePA). CMS can encourage payers to use national networks by financially incentivizing payers to use those networks for multiple use cases, including prior authorization, electronic evidence collection for reimbursement claims processing, membership management, quality, risk stratification and more.

   CMS can encourage payers to modernize by mandating HL7 DaVinci FHIR-based Payer-to-Payer and Payer-to-Provider Access APIs. The CMS could offer incentives for early adoption and production use of TEFCA and other national health data sharing networks. Payers covering 34 states have joined eHealth Exchange to facilitate payer-provider data exchange within the eHealth Exchange network, although these payers have not yet elected to also exchange with providers in different TEFCA networks (due to the current governance and potential fee structure for non-Treatment queries, as well as the current design of "Facilitated FHIR"). Technical assistance and developer sandboxes, like those eHealth Exchange provides, would accelerate implementation. These steps would enhance data sharing and reduce the time and expense of care and payment coordination.

3. Public health has been making progress, in terms of increased connectivity, but is hampered by being largely bound to the 1970s HL7 v2 protocol. The CMS should provide financial support for updating existing public health use cases to FHIR. Support could be in the form of grants to data sources to adopt FHIR bi-directional exchange. FHIR Bulk data operations are now becoming widely supported by data holders (such as hospital EHRs). We believe eHealth Exchange is the first national network transacting payer provider exchange using HL7 Bulk Data (FHIR) in production. This, and targeted push and pull of FHIR data, can provide a wealth of information to public health while modernizing and standardizing on FHIR.

The CMS could mandate that TEFCA data holders MUST respond to Public Health, Insurance, and Research use cases. Currently, TEFCA allows for these three use cases, but TEFCA does not require them. Hence, to our knowledge, no data holder on TEFCA will respond to these eXchange Purposes other than as detailed in *Mini Case Studies*. This is a lost opportunity. By mandating responses to Public Health, Insurance, and Research use cases, public health organizations and payers could start to leverage the nation's investment in TEFCA to and re-use the existing TEFCA network for more, high priority, use cases. Below we provide a case study of how two federal agencies are leveraging FHIR, and national networks, to modernize adverse reaction case follow-ups.

4. Increased use of Bulk FHIR, supported by eHealth Exchange today in production, would enhance population health analytics, quality reporting, and public health surveillance. Bulk FHIR enables efficient data acquisition of all authorized patients in specified patient populations via a single API call, reducing provider and public health burden. eHealth Exchange's use of Bulk FHIR for public health use cases demonstrate Bulk FHIR's viability. The CMS could mandate adoption of Bulk FHIR, honoring appropriate Minimal Data Set constraints, to improve national data acquisition cost efficiency. Bulk FHIR data holders' capabilities (facility-specific query identifiers) could be added to a national directory to facilitate interoperability. Below is a case study on how eHealth Exchange is leveraging Bulk FHIR.

5. Currently, organizations involved with safety net operations, such as those assisting with housing instability, food insecurity, and other similar issues, are fragmented and non-interoperable. Existing national networks can be leveraged to onboard this important and largely disconnected group of actors. The CMS should convene stakeholders, help achieve broad consensus and then incentivize these safety net organizations to participate in national networks with a single, FHIR-based, unified approach to bi-directional interoperability with payers and providers. This will multiply the effectiveness of clinical care providers and payers by allowing them to quickly and cost-effectively intervene, when needed, to help ensure patients have a supportive environment allowing them to adhere to best practices and recover from, and prevent, illnesses.

6. Patient authorization assertions using HIPAA Authorization forms for non-Treatment, Payment, and HealthCare Operations releases are largely an unintended barrier to interoperability. These authorizations today typically

leverage a manual process using non-machine-readable HIPAA Authorization forms. FHIR provides for computable consent expressions (see examples here: https://build.fhir.org/consent-examples.html) which could accommodate HIPAA Authorization forms for non-Treatment, Payment, and HealthCare Operations releases. The CMS could consider facilitating working groups under IHE USA and HL7 to pilot, refine, and then financially incentivize production use of computable consent nationally. This could be broadened to computable patient authorization assertions to enable Research and other non-HIPAA covered exchanges such as for disability payments. It could be further broadened to Data Segmentation for Privacy (DS4P) to enable EHRs and other actors to tag certain data elements as being subject to special privacy and data-sharing rules.

7. FHIR based exchange, while improving, has gaps. These include:

    1. A standards-based mechanism to unambiguously route FHIR messages between actors via an intermediary (such as a national network).

    2. A standards-based method to express FHIR message metadata such as the purpose of the transaction (treatment, public health, payment, etc.), the originating person or system, the role of the originating person in the requesting organization, and more.

    3. Security models are currently not viable at scale for either patient-based trust, or organization-based trust. The HL7 FAST workgroup has made sporadic progress on these topics over the last 3 years but, unfortunately, the standards are not complete yet. The CMS could convene, and if needed fund, work under HL7 FAST to pilot and refine these standards until they are ready for production use. See our response to the question TD-3.c

    4. FHIR push (from external sources to EHRs and HIEs) should be supported to allow authorized exchange partners, labs, medical IoT, and patient contributed data in FHIR format. Those “pushed” data elements could be stored in an “inbox” for review by authorized individuals before incorporation into the medical record to support proper physician review. Or they could be tagged as patient-contributed data so provenance is clear. Currently the vast majority of EHRs only allow inbound data in HL7 V2 format. This prevents implementation of important use cases. Inbound patient-worn devices such as smart watches now include FDA approved features such as heart rhythms deviations, temperature, blood oxygen, and more. Referrals from a primary care provider to patients cannot be done using FHIR because of restrictions in acceptance of inbound FHIR data. Many other, clinically valuable use cases are currently unimplementable due to this restriction. The result: These data

are either lost, or they are incorporated into the medical record incompletely, via costly manual data entry.

Sample FHIR push use cases:

1. **Home Health:**
    1. Data: Nursing assessments, medication administration records, physical/occupational therapy notes.
    2. FHIR Block: Fragmented systems and reluctance to integrate these often-smaller providers.
    3. Impact: Gaps in care coordination and potential for medication errors or adverse events.
2. **School Health Records:**
    1. Data: Immunizations, allergies, chronic conditions, developmental assessments.
    2. FHIR Block: Integrating data from separate school systems.
    3. Impact: Lack of a complete health picture for pediatric patients.
3. **Emergency Department (ED) Follow-Up Coordination:**
    1. Data: ED visit summaries, discharge instructions, follow-up appointments.
    2. FHIR Block: Difficult to push these back into the patient PCP EHR without manual intervention or complex HL7 v2 interfaces.
    3. Impact: Fragmented care, lack of continuity between ED and primary care.
4. **Specialty Clinic/Lab Data Integration:**
    1. Data: Results from independent labs (genetic testing, pathology), specialized clinics (cardiology, oncology).
    2. FHIR Block: EHRs don't accept FHIR test results or clinical notes from external sources. This results in manual data entry or faxes.
    3. Impact: Incomplete patient history, potential for errors, expensive manual fax and telephone communications, delays in diagnosis/treatment.
5. **Wearable Device Data (Beyond Basic Tracking):**
    1. Data: ECG data from smartwatches, blood glucose trends from advanced CGMs, sleep apnea metrics from sleep trackers.
    2. FHIR Block: Limited FHIR support for advanced device data.
    3. Impact: Underutilization of valuable patient-generated health data.
6. **Long-Term Care Facility (LTC) Data:**
    1. Data: Nursing assessments, medication administration records, incident reports.
    2. FHIR Block: Integration with LTC facilities is often disconnected from the healthcare ecosystem.
    3. Impact: Lack of a comprehensive view of patients receiving long-term care.
7. **Closed-Loop Referrals:**
    1. Data: Referrals orders to specialists and results back to PCPs
    2. FHIR Block: Lack of inbound FHIR messages to EHRs

3. Impact: “Closed Loop Referrals” use older standards or manual communications like faxes and telephone calls which are expensive, incomplete, slow, and error prone.

8. Currently, there are at least EIGHT competing standards for healthcare and payment data exchange. These should be leveraged for a reasonable amount of time while each of these various domains is incentivized to move to FHIR transactions.

   Standards such as NEMSIS for ambulance data exchange, NIEM for the Department of Justice, X12 for insurance, HL7 v2 for legacy labs and hospitals and public health reporting, IHE SOAP/SAML/XML for many national networks, DirectTrust secure email, HL7 v3 for national networks, and now FHIR. Additionally, some vendor-driven national networks use proprietary APIs. It is likely in the national best interest for the CMS to set a vision, and provide financial incentives, for a single standard, such as HL7 FHIR r4 (and above) with IHE FHIR standards for all these use cases. This ideally is a deliberate, nonpartisan, long-term project with a broadly approved timeline that leverages the interoperability deployed today, while giving us all a clear target for the future.

9. eHealth Exchange cannot stress enough the importance of non-profit self-sustaining HIEs, such as eHealth Exchange who have never received a grant, who are committed to not monetizing the data we are trusted to protect. This has resulted in eHealth Exchange being federated and thus storing no clinical data in a centralized data repository. This results in a reduced cybersecurity attack surface area and helps keep patient data safer. The CMS could provide an environment that leverages these national networks that play a vital role in connecting actors, and in innovating new approaches to connect other actors such as safety net and payers.

10. eHealth Exchange, as detailed below in *Mini Case Studies*, has deployed national-scale FHIR. This pioneering work, as the first national network to enter into production with FHIR, can provide specific examples that prove-out the viability of Networked FHIR and inform CMS. eHealth Exchange supports the following FHIR-based use cases in production:

    1. **Patient Access API:** (TEFCA Individual Access) Supports individual queries for patient data, used for care coordination, compliant with CMS rules.

2. **Directory API:** Facilitates provider lookup, searchable by NPI, used for all partner-to-partner communications, including across the approximately 50 networks eHealth Exchange is connected to.
3. **HL7 Bulk Data (FHIR):** Supports population-level queries with Group ID filtering, used for analytics and population health.
4. **SMART on FHIR:** Supports EHR-launched and standalone apps, with HL7 FAST enhanced OAuth 2.0 deployment processes. Leverages both FHIR and IHE XCA APIs.
5. **Payer HL7 DaVinci** transactions including prior authorization workflows.
6. **Targeted query/retrieve** for both treatment and public health (adverse event case follow up).
7. **Push** suspected adverse events to public health.

11. eHealth Exchange has recently demonstrated the use of national networks for the collection and exchange of quality measurements, such as HEDIS data elements. A case study of this important use case is provided below in *Mini Case Studies*. CMS could leverage national networks to capture and exchange quality measurement data via financial incentives to reporting organizations.

12. The CMS specifically asks what alternatives to exchange exist outside of TEFCA. eHealth Exchange, a non-profit founded for the public good as a 501(c)(3) by the HHS/CMS, is the oldest and largest health data sharing network in the country, to our knowledge. It has never accepted a government or private grant, is financially sustainable, and it presents a viable national network that continues to connect to all other available regional, vendor-lead, specialty, and national networks.

    Since the TEFCA framework is not funded by federal government grants and has no publicized sustainability model, some question TEFCA's future. Self-funded eHealth Exchange has demonstrated value and sustainability, vendor neutrality, and a history of innovating while freely sharing its success and industry knowledge via IHE, HL7, and additional forums. eHealth Exchange has also demonstrated a consistent willingness to embrace and connect with all other networks for the better good. eHealth Exchange serves as a compelling example for national-scale connectivity for the nation just as it did in 2006 as an HHS initiative.

Mini Case Studies

Below we share several production deployments that we believe may help inform the CMS with some specific examples, including both successes and remaining challenges for the industry to overcome. More information on each of these case studies is available from eHealth Exchange. Note that although we mention several federal agencies below, we are not attempting to represent their perspectives or communicate on their behalf; this is the eHealth Exchange perspective on these projects.

Case Study #1: APHL Public Health Electronic Case Reporting (eCR)

Challenge: A public health organization that receives reportable conditions (such as Tuberculosis) notifications from health systems sought to avoid multiple point-to-point connections from each health system in the nation. **Actions:** Through their existing connection to the APHL AIMS platform, the public health organization was able to leverage the eHealth Exchange as a single technical connection to the vast majority of health systems in the US. With eHealth Exchange's Hub, each health system connects to the Hub. The Hub receives messages from each reporting health system, and then securely routes it to the AIMS Platform which distributes the electronic initial case report to the appropriate public health jurisdiction. **Results:** Regarding the data contributors, many health systems, onboarding to eHealth exchange and sending eCR via APHL/eHealth Exchange may be the first time they are able to report electronically; this is often a move from manual reporting, which is a major time-savings and improvement in data quality, for both the health system and the public health organization. The organization was able to reduce costs by decommissioning legacy reporting mechanisms and maintaining a single connection for reportable conditions. As they move towards adoption of FHIR standards, specifically Networked FHIR, they will be able to leverage a single connection with the eHealth Exchange for every FHIR pushed notifiable condition transaction. This will allow them to keep their existing infrastructure in place as they move towards FHIR based exchange at a pace that can be supported by their exchange partners. Since the eHealth Exchange Hub maintains no centralized clinical data repository, the public health organization has more confidence in the cybersecurity of this deployed architecture. For More Information: 1. COVID Go Live: https://ehealthexchange.org/ehealth-exchange-launches-electronic-case-reports-ecr-nationwide/ 2. Curent Sites: https://www.cdc.gov/ecr/php/healthcare-facilities/index.html : 3. TEFCA: https://www.cdcfoundation.org/TEFCA-eCR-Query-Hartsell?inline#:~:text=What%20is%20Electronic%20Case%20Reporting%20(eCR)?%20The,Agreement%20(TEFCA)%20Qualified%20Health%20Information%20Network%20(QHIN).

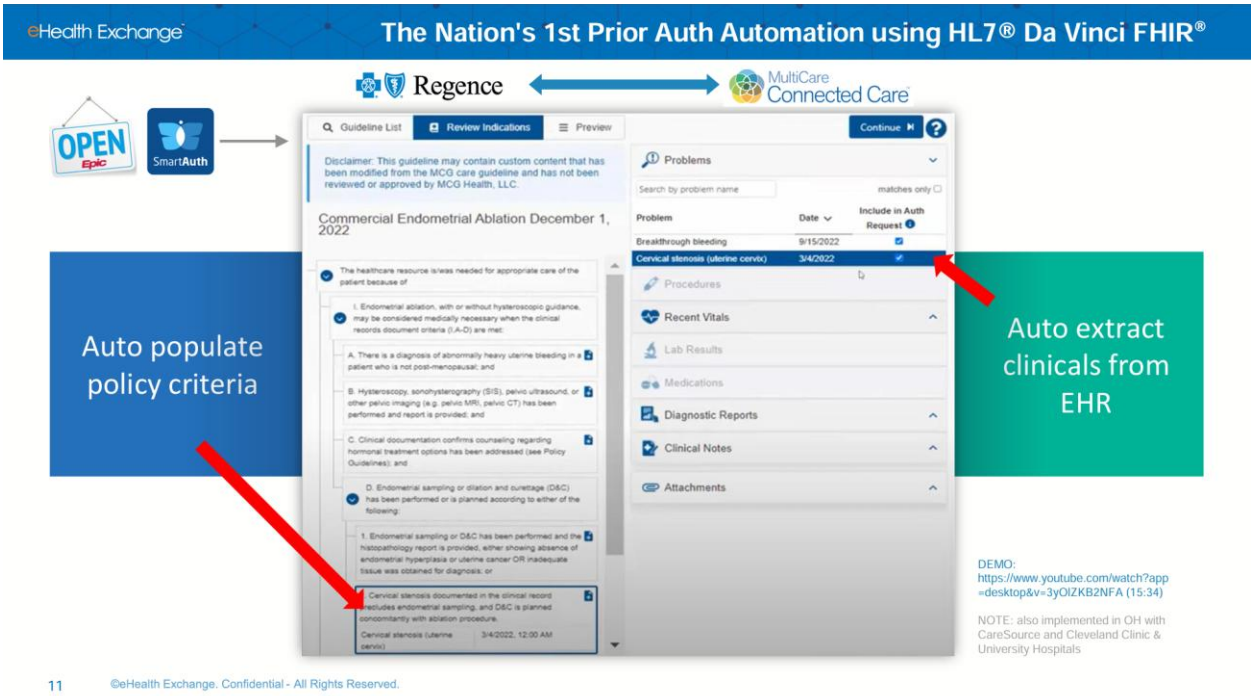
Case Study #2: Federal Agency Use of eHealth Exchange and FHIR for Population Health Bulk Queries, Adverse Reactions Notifications, and Case Follow-up

Challenge: A federal agency was seeking to implement three use cases: a) replace an electronic fax system (from another federal agency) sending suspected regulated product adverse reactions with modern standards, b) to perform bulk queries to retrieve multiple patient records with one query where that patient population was defined by specific observable characteristics (phenotypes), and c) perform targeted queries to follow up with suspected adverse event reports. **Actions:** a) Both agencies connected their FHIR end points to the eHealth Exchange Production FHIR Hub. Since the eHealth Exchange DURSA multi-party legal agreement was in place, the agencies were able to exchange using that legal and trust framework. b) The federal agency and the eHealth Exchange worked with approximately 20 data holders to implement FHIR bulk queries at each data holder location. c) Each data holder in this production data pilot was individually recruited and onboarded with a single Networked FHIR connection to the Hub. Flexible security models were deployed. **Results:** Respectively: a) Both agencies were able to successfully implement FHIR push transactions of patient, medication and adverse event FHIR resources. b) The agency was able to retrieve entire patient sub-populations from data holders with a single FHIR API call, honoring applicable Minimal Data Set constraints. c) Targeted FHIR queries were successfully implemented resulting in that agency being able to retrieve data in seconds to follow up on suspected adverse event cases. Overall, an extra benefit realized by all participants in all three of these sub-use cases, due to their use of Networked FHIR, was that their organizations were now ready for additional FHIR transactions with other eHealth Exchange Participants. **Remaining Challenges:** The standards-based FHIR approach has several challenges: security model, routing, and metadata.

Case Study #3: Prior Authorization at Scale: A National First

Challenge: Prior authorization transactions often require manual effort, are expensive, result in care delays, do not use modern standards, and do not scale. The HL7 DaVinci standards and CMS 0057 rule provide a pathway to eliminating this burden, but adoption is slow, and the industry is entrenched in legacy methods. The industry lacks the necessary leadership to truly automate Prior Authorization processes.
Actions: A group of organizations including a technology provider, multiple payers, and multiple provider organizations joined an informal coalition called Project Trebuchet, led by the eHealth Exchange, to implement and “launch” into production, the nation’s first FHIR-based prior authorization transactions. **Results:** Providers and payers enabled and tested Prior Authorization workflows and can now use SMART on FHIR and deploy prior authorization electronic workflows without EHR support. The use of SMART on FHIR accelerated electronic Prior Authorization (ePA) timelines, allowing providers to select their own ePA experience instead of being dependent upon (potentially non-existent) EHR built-in ePA capabilities. The eHealth Exchange Hub implemented a

"prior authorization router" that identifies the payor for each prior authorization request and automatically submits the correct prior authorization end point automatically. **Remaining Challenges**: There are many challenges remaining to scaling Prior Authorization deployment in the US; the CMS could play a crucial role in resolving these issues by engaging. Payers and providers are not fully aligned on how the ePA processes should function. Some vendors continue to play gatekeeper roles and impose arbitrary limitations on provider organization data access. TEFCA does not yet have a defined pathway for ePA connectivity between parties and lacks specific provider directory components to support payer connectivity. **For More Information:** Please see the public whitepaper at the following URL. It contains detailed case studies, technical information, and project diagrams. https://ehealthexchange.org/wp-content/uploads/2024/10/eHealthExchange_DaVinci_BurdenReduction_TechSpecs_ImplementationProcesses_9.4.2024.pdf and https://ehealthexchange.org/ehealth-exchange-honored-with-klas-points-of-light-award-for-advancing-automated-prior-authorization-via-networked-fhir-service/

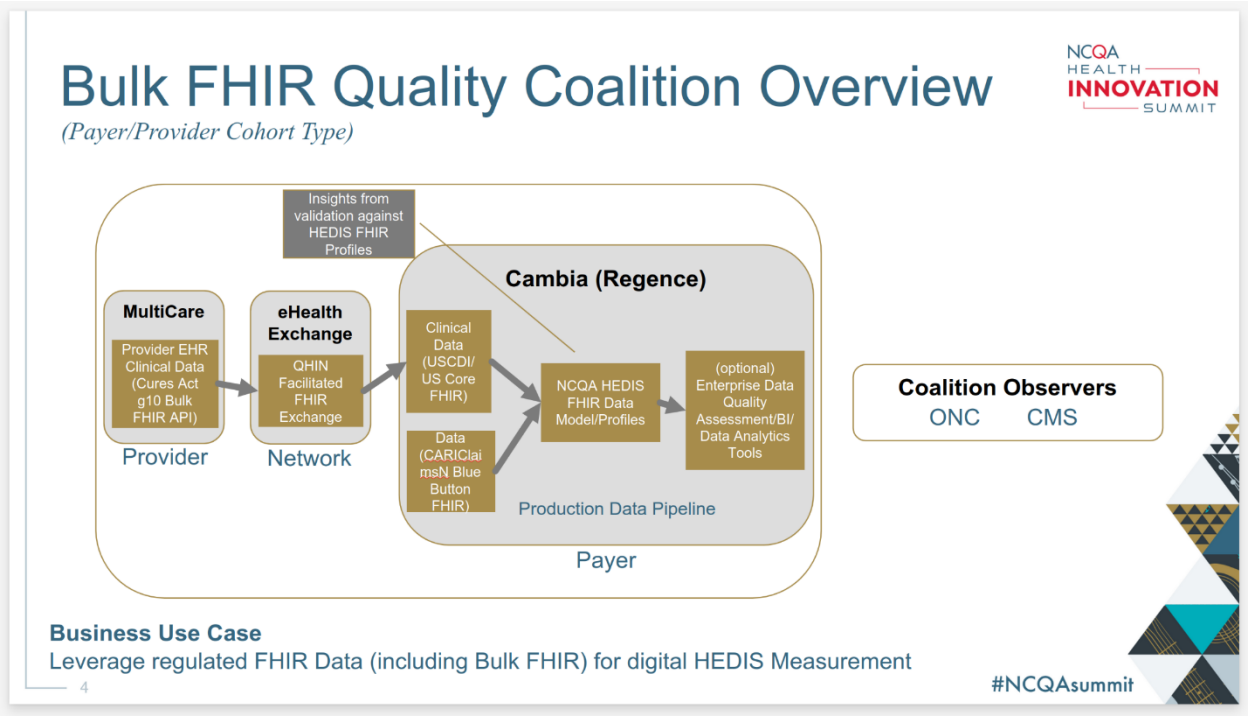


The nation's 1st Prior Authorization Automation Solution, Image from Public White Paper

Case Study #4: Use of the eHealth Exchange and Bulk FHIR for Digital HEDIS Measurement

Challenge: The objective for the inaugural cohort is to exchange the clinical data needed for HEDIS measurement for 5,000 to 10,000 patients in a matter of minutes. The cohort will share insights and analytics on: Performance of regulated Bulk FHIR APIs, Quality of Bulk FHIR data, Integrity, Conformance, Completeness, Plausibility,

The content for following HEDIS Measures can be used for deeper data quality insights: Controlling High Blood Pressure (CBP), Depression Screening and Follow-Up for Adolescents and Adults (DSFE), Hemoglobin A1c Control for Patients With Diabetes (HBD). **Actions:** eHealth Exchange enabled connectivity between cohort participants using the HL7 Bulk Data pattern. Data retrieval processes between cohort participants flowed smoothly, allowing Cambia to obtain the required FHIR data for their HEDIS quality measurement efforts. **Results:** Cambia (Health Plan) and eHealth Exchange (TEFCA QHIN) and MultiCare-enabled production data flows using the HL7 Bulk Data standard. **Remaining Challenges and Recommendations:** Sensitive Data Processing: Awaiting updates from the data holder to process sensitive data from FHIR APIs effectively. HEDIS Quality Measurement: Limited support from HEDIS vendors for FHIR data. CMS should mandate HEDIS measurement calculation vendors ingest FHIR data for HEDIS measures and promote recognition of data quality from validated § 170.315(g)(10) FHIR APIs without re-certification.



Bulk FHIR Quality Collation, Image from our public presentation at the NCQA Health Innovation Summit

Networked FHIR – Connect Once, Exchange with All

CMS should leverage existing national networks, like eHealth Exchange and TEFCA, in a proven hub-and-spoke model known as "Networked FHIR" instead of deploying point-to-point FHIR connections.

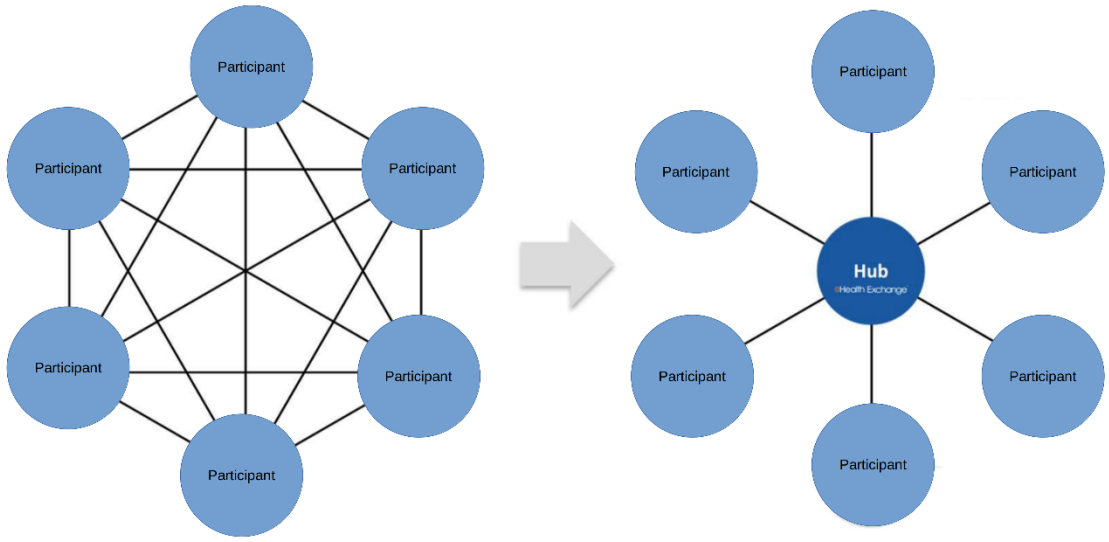
Networked FHIR could allow patients, providers, agencies and public health to leverage single connections for multiple purposes (measurement data capture, quality data transmission, clinical data sharing, public health use cases, and much more).

Networked FHIR is a scalable approach to health data exchange in which Participants, such as CMS, connect once to a trusted national Hub and immediately gain secure bidirectional access to every other Participant's FHIR APIs. **The burden is significantly reduced for every Participant by implementing a nationwide FHIR hub-and-spoke model such as eHealth Exchange's existing, live HL7 DaVinci compliant solution.** Instead of manually creating one-off connections with thousands of Participants, Participants send their FHIR API calls to the shared Hub, where common governance, authentication, auditing, security, and version management are handled for all. This model reduces cost and complexity, scales effortlessly as new members join, and keeps every Participant current with the latest standards, creating a continuously expanding network of interoperability.

Referring to the diagram below, the long-term operational costs are represented by the connections. In peer-to-peer FHIR, the number of connections grows exponentially. In a hub-and-spoke model, the number of connections remains ONE single connection.

Point-to-Point FHIR
(expensive up-front costs, expensive operational costs, less secure, and not scalable)

Networked FHIR
(exponentially lower up-front costs, single connection to maintain and secure, and scalable)



Point-to-Point FHIR v Networked FHIR

In other industries, there is a long history of business-to-business data flows being routed through a central hub, while direct point-to-point exchange has remained rare and cumbersome. Financial clearinghouses, travel reservation systems, and supply chain networks connect thousands of firms through one shared backbone. Healthcare follows the same pattern. National networks such as eHealth Exchange have proven that a single shared Hub can give hospitals, payers, government agencies, and public-health programs near instantaneous, secure, access to one another's clinical and payment data. This Hub model, Networked FHIR, allows an organization to connect once and instantly reach a growing network of partners, rather than negotiating, manually securing, testing, implementing, and maintaining an expensive new FHIR connection for every new exchange partner.

Attempts to build point-to-point FHIR connections still appear, but they are expensive and impractical. And they only work for the two parties that funded that single connection. Each new connection repeats the legal reviews, security audits, and software customization of the last one. Few providers can spare the time or money to sustain dozens of such bespoke FHIR connections, so these efforts rarely scale beyond a limited small group of Participants.

More significantly, eHealth Exchange, which was a point-to-point network when originally founded, discovered that the long-term expense to Participants was not in the up-front costs of building the point-to-point connections, it was in the long-term operational maintenance of each connection. Each connection fails. Time and manual effort are required to diagnose and fix each connection failure. Each connection also has to be upgraded to incorporate new security standards, bug fixes, etc. This is also a risky, and expensive, manual expense per connection. Finally, each connection introduces additional cybersecurity risks. Over time, this has simply proven to be impractical, even for large organizations.

In contrast, Networked FHIR's centralized Hub offers economies of scale, shared operational expenses, and uniform security controls. It helps every Participant move faster when standards evolve, because one upgrade at the Hub lifts the whole community. As more Participants join, the value of membership in the Networked FHIR ecosystem grows for all Participants, reinforcing the network and drawing in new use cases such as prior authorization, public-health reporting, and value-based-care analytics. Very importantly, another implication of Networked FHIR is that there is only a single connection to defend against cyber-attacks; With point-to-point FHIR exchange, there are multiple connections to defend.

The security model of Networked FHIR is critical, as well; it allows system level trust models which can be automated at scale.

The CMS should leverage the large number of benefits Networked FHIR offers, and the CMS should leverage the DURSA legal agreement. The alternative, a national point-to-

point approach, has proven to be a costly mistake. Using the eHealth Exchange's Networked FHIR and DURSA would allow the CMS to have a single connection to eHealth Exchange to gather modern esMD-like quality data and evidence data from the rest of the nation. Providers need a single solution for quality reporting and eHealth Exchange already provided ESRD quality reporting for small independent dialysis centers to the CMS.

For the CMS the lesson is hopefully clear; Most successful B2B ecosystems rely on a trusted hub, not a patchwork of direct connections. By embracing Networked FHIR the CMS can align with the dominant pattern of digital B2B transactions, reduce duplication, boost security, and accelerate nationwide interoperability.

Specific CMS RFI Questions and Responses

Re: CMS RFI: RIN 0938-AV68, Due June 16, 2025, [CMS-0042-NC]
https://www.federalregister.gov/documents/2025/05/16/2025-08701/request-for-information-health-technology-ecosystem

PC-4. a. What apps should exist but do not yet? Why do you believe they do not exist yet?

eHealth Exchange envisions apps that combine real-time clinical data, claims, and SDOH insights to provide holistic care navigation, such as matching Medicare beneficiaries to providers based on quality, cost, and other criteria important to the patient. These apps do not yet exist due to challenges in accessing non-clinical data (e.g., housing, transportation), inconsistent FHIR endpoint availability, and developer hesitancy due to complex regulatory requirements like HIPAA. They also do not exist due to the lack of an authoritative national directory. CMS could incentivize development by expanding API access to SDOH data and creation of a national directory.

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

eHealth Exchange's network makes clinical data, such as lab results, medications, allergies, and care summaries, readily available to patients and caregivers via FHIR-based Patient Access APIs. These data types are valuable for managing chronic conditions, ensuring medication adherence, and coordinating care across providers. Unfortunately, FHIR based connections still require one-off manual coordination for each connection to each data holder, at this time. The CMS should encourage completion and adoption of scalable FHIR security standards.

PC-8. a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

Valuable but hard-to-access data includes real-time encounter notes, SDOH (e.g., housing, food security), and longitudinal claims data. App developers face challenges due to inconsistent provider adoption of FHIR endpoints, non-standardized data formats (e.g., scanned documents), and restrictive consent policies. eHealth Exchange mitigates these issues by supporting IHE XCA and FHIR standards, but broader CMS mandates for USCDI compliance and information blocking enforcement would enhance access for developers and patients alike.

PC-9. a. What difficulties are there in accessing or utilizing these data sources today?

Difficulties include technical barriers like authentication challenges and varying consent requirements across states. The CMS could assist by requiring a scalable authentication mechanism such as those advocated by HL7's FAST work group, and by requiring the consent expressions to be computable, interoperable, and electronically available (e.g. the FHIR Consent resource). Networked FHIR could assist in the rapid deployment of these capabilities.

PC-10. c. What use cases could have a significant impact if implemented through TEFCA?

Public Health, Insurance, and Research use cases would be highly impactful to the country. Specifically, making responses mandatory for these three use cases, while honoring Minimal Data Set constraints. For example, each person in the country could have their fully de-identified MDS data contributed to a virtual clinical trial, as appropriate and authorized under appropriate authorities. This would solve some of the current problems with clinical trials (recruiting and lack of representative demographic groups). It also could enable post-market efficacy analysis of regulated products. Finally, electronic study recruitment would result in significant national cost savings compared to the current expensive and manual process.

Regarding CMS-specific operations: Using the Networked FHIR and DURSA would allow the CMS to have a single connection to eHealth Exchange to gather modern esMD, ESRD and HEDIS data and evidence data from the rest of the nation.

PC-10. e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?

See *Broad Themes, Topic #7.4 related to FHIR Push.*

PC-10. f. Are there redundant standards, protocols, or channels that should be consolidated?

Currently, there are at least EIGHT competing standards for healthcare and payment data exchange. These should be leveraged for a reasonable amount of time while each of these various domains is incentivized to move to FHIR transactions.

Standards such as NEMSIS for ambulance data exchange, NIEM for the Department of Justice, X12 for insurance, HL7 v2 for legacy labs and hospitals and public health reporting, IHE SOAP/SAML/XML for many national networks, DirectTrust secure email, HL7 v3 for national networks, and now FHIR. Additionally, some vendor-driven national networks use proprietary APIs. It is likely in the national best interest for the CMS to set a vision, and provide financial incentives, for a single standard, such as HL7 FHIR r4 (and above) with IHE FHIR standards for all these use cases. This ideally is a deliberate, nonpartisan, long-term project with a broadly approved timeline that leverages the interoperability deployed today, while giving us all a clear target for the future.

PC-10. g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

While alternatives like vendor networks and regional HIEs exist, they lack nationwide scope and a standardized trust framework. In eHealth Exchange's role as a QHIN, we help bridge these gaps. We currently connect 75% of hospitals, 5 federal agencies, 58 regional and state HIEs, 64 public health jurisdictions, 70K medical groups, 85% of the nation's dialysis centers, and 34 states with payer exchange. Without TEFCA, fragmentation persists due to inconsistent standards and consent models. TEFCA, supported by our network, is the most robust solution for widespread, secure patient access.

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

eHealth Exchange, as a network of 58 regional and state HIEs, advances patient access by enabling the rest of the country to query for comprehensive health records localized to a region. Some eHealth Exchange local HIEs provide centralized clinical data repositories for their region with value-added services such as community health metrics, provider alerts, risk stratification, reporting, and other functions. Regional HIEs can also provide additional services such as notification upon patient admission to emergency departments, and partnerships with state public health authorities. CMS should encourage these regional HIEs to adopt FHIR-based bidirectional standards such as Bulk FHIR, push, and query/retrieve.

PC-11. b. What changes would you suggest?

eHealth Exchange suggests CMS provide financial incentives for regional HIEs to adopt FHIR R4 (and above) universally to enhance data standardization and support real-time access. CMS could incentivize HIEs to integrate SDOH and public health data, expanding their utility. Additionally, streamlining consent frameworks across HIEs would reduce barriers to data sharing.

HIEs should also accept inbound FHIR Push messages. See *Broad Themes, Topic #7.*

PC-11. c. Are there particular examples of high-performing HIE models that you believe should be propagated across markets?

eHealth Exchange cannot stress enough the importance of non-profit, self-sustaining, HIEs such as eHealth Exchange who are committed to not monetizing the patient data we are trusted to protect. HIEs committed to offering near real time data, providing complete and accurate patient records, and dedicated to not-monetizing patient data, are characteristics of high-performing HIE's.

eHealth Exchange has no centralized data repository of clinical data. This is important as it helps us avoid being an attractive target for cybercriminals. Also, the eHealth Exchange does

not resell patient data. That is critical to help establish trust from data holders and patients alike. Trust is a prerequisite for data exchange.

PC-11. d. What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?

eHealth Exchange cannot stress enough the importance of non-profit, self-sustaining, HIEs such as eHealth Exchange who are committed to not monetizing the patient data we are trusted to protect. HIEs committed to offering near real time data, providing complete and accurate patient records, and dedicated to not-monetizing patient data, are characteristics of high-performing HIE's.

PR-5. Which of the following FHIR APIs and capabilities do you already support or utilize in your provider organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints:

eHealth Exchange supports multiple FHIR APIs for participants, including:

- **Patient Access API:** (TEFCA Individual Access) Supports individual queries for patient data, used for care coordination, compliant with CMS rules.
- **Provider Directory API**: Facilitates organization lookup, searchable by NPI, used for all partner-to-partner communications including across the different networks eHealth Exchange supports.
- **Bulk FHIR:** Supports population-level queries with Group ID filtering, used for HEDIS analytics and public health. Could be used for study recruitment in the future.
- **SMART on FHIR:** Supports EHR-launched and standalone apps, with HL7 FAST enhanced OAuth 2.0 deployment processes. We utilize both FHIR endpoints and IHE XCA. Used for electronic Prior Authorizations.
- **Targeted Query/Retrieve:** Supports standard FHIR query and retrieve operations. Used for treatment use cases, public health, and could potentially be used for insurance, research, and more if TEFCA mandated responses to these use cases.
- **Push:** Supports pushing FHIR data from the data holder to the recipient. Potential use cases include reportable condition notifications, encounter alerts, subscripted alerts, closed-loop-referrals, and adverse events notifications. See *Broad Themes, Topic #7.*

PR-6. Is TEFCA currently helping to advance provider access to health information?

PR-6. a. Please provide specific examples.

Please see the *Mini Case Studies* section of this document.

PR-6. b. What changes would you suggest?

The CMS should mandate that TEFCA data holders respond to Public Health, Insurance, and Research use cases. Currently, TEFCA allows for these use cases, but TEFCA does not require them. Hence, to our knowledge, no data holder on TEFCA will respond to Public Health, Insurance, or Research eXchange Purposes other than as detailed in *Mini Case Studies*. This is a lost opportunity. The CMS should mandate responses for these three purposes to leverage the nation's investment in TEFCA to and re-use the existing TEFCA network for these high priority uses.

TEFCA should deploy Networked FHIR instead of non-scalable point-to-point FHIR. See the section *Networked FHIR* in this document for more information.

TEFCA's use of FHIR has the potential to advance provider access by enabling more efficient, secure, private, and standardized data exchange across networks based on correctly-scoped healthcare FHIR data objects instead of broadly-scoped C-CDA documents. Document exchange is valuable, and effective, but it also can result in sharing more information than is needed. For example, many specialists are seeking only those data relevant to them, but document-based exchange normally results in additional information being transmitted. Another example is disaster response where providers are mostly focused on current problems, current medications, and drug allergies. A complete C-CDA document has much more information that is not used in a disaster response treatment scenario. With FHIR, the requester can be more specific and only retrieve those clinical data elements of interest. This is privacy-protecting and more efficient for all actors. It should also result in cost savings due to much more efficient data sharing (smaller transaction sizes).

The CMS should leverage Networked FHIR offers, and the CMS should leverage the DURSA legal agreement. The alternative, a national point-to-point approach, has proven to be a costly mistake. Using the eHealth Exchange's Networked FHIR and DURSA would allow the CMS to have a single connection to eHealth Exchange to gather modern esMD-like quality data and evidence data from the rest of the nation. Providers need a single solution for quality reporting and eHealth Exchange already provided ESRD quality reporting for small independent dialysis centers to the CMS.

PR-6. c. What other options are available outside of TEFCA?

The eHealth Exchange, a non-profit founded for the public good as a 501(c)(3) by the HHS, is the oldest and largest health data sharing network in the country, to our knowledge. It has never accepted a federal agency or private grant, is financially sustainable, and it presents a viable national network that continues to connect to all other available regional, vendor-lead, specialty, and national networks. eHealth Exchange has no centralized data repository of

clinical data. This is important as it helps us avoid being an attractive target for cybercriminals. Also, the eHealth Exchange does not resell patient data. That is critical to help establish trust from data holders and patients alike.

In addition, there are approximately 50 other (largely vendor-lead) networks at this point; a listing can be found here: https://carequality.org/carequality-interoperability-framework-adopters/ and 58 regional HIEs, a list of which may be found here: https://ehealthexchange.org/participants/?participant_type=hie.

Finally, there are many specialty networks and data sources that remain largely non-interoperable such as state PDMPs, cancer and other disease-specific registries, immunization registries, research networks, and more. The CMS should mandate that all healthcare specialty networks and data sources move towards FHIR and connect to TEFCA and other national networks. Otherwise, these valuable data remain isolated and underutilized.

PR-6. d. Are there redundant standards, protocols or channels or both that could be consolidated?

Currently, there are at least EIGHT competing standards for healthcare and payment data exchange. These should be leveraged for a reasonable amount of time while each of these various domains is incentivized to move to FHIR transactions.

Standards such as NEMSIS for ambulance data exchange, NIEM for the Department of Justice, X12 for insurance, HL7 v2 for legacy labs and hospitals and public health reporting, IHE SOAP/SAML/XML for many national networks, DirectTrust secure email, HL7 v3 for national networks, and now FHIR. Additionally, some vendor-driven national networks use proprietary APIs. It is likely in the national best interest for the CMS to set a vision, and provide financial incentives, for a single standard, such as HL7 FHIR r4 (and above) with IHE FHIR standards for all these use cases. This ideally is a deliberate, nonpartisan, long-term project with a broadly approved timeline that leverages the interoperability deployed today, while giving us all a clear target for the future.

Regarding redundant channels, unfortunately there are a vast number of alternative approaches that reduce our national efficiency. TEFCA, and eHealth Exchange, Networked FHIR offers a compelling alternative: One connection to the country. The CMS should leverage the large number of benefits Networked FHIR offers, and the CMS should leverage the DURSA legal agreement. The alternative, a national point-to-point approach, has proven to be a costly mistake. Using the eHealth Exchange's Networked FHIR and DURSA would allow the CMS and all other Participants to utilize a single connection to eHealth Exchange to share data with the rest of the nation.

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

CMS can support providers by mandating FHIR R4 (and above) endpoints for reading and writing clinical data from and to EHRs, enforcing information blocking rules, and providing financial incentives for data sharing, aligning with eHealth Exchange and TEFCA standards. To mitigate burden, CMS could integrate interoperability metrics into existing quality programs (e.g., MIPS), rewarding providers for using FHIR APIs. Reusing workflows like quality reporting, supported by Bulk FHIR, would streamline data sharing, leveraging the eHealth Exchange (and other national networks') to reduce provider effort.

For HEDIS, CMS should mandate HEDIS measurement calculation vendors ingest FHIR data for HEDIS measures and promote recognition of data quality from validated § 170.315(g)(10) FHIR APIs without re-certification. See *Mini Case Studies / Use of the eHealth Exchange and Bulk FHIR for Digital HEDIS Measurement*.

To avoid providers needing to pay for, deploy, secure, and manage multiple connections, the CMS should support the use of Networked FHIR hence allowing providers to only have to maintain a single connection to interoperate with national networks. Please see *Networked FHIR* for more information. Using Networked FHIR could allow patients, providers, agencies and public health to leverage single connections for multiple purposes (measurement data capture, quality data transmission, clinical data sharing, public health use cases, and much more).

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

The CMS ASTP/ONC should recognize a new category of information blocking: that of incomplete information sharing. Some data holders, for example, only share a limited subset of all of their data on a given patient. The CMS should mandate that ALL legally allowable, and patient approved, clinical data be available for retrieval via approved participants in national networks.

PR-13. For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?

Laboratories and imaging centers are two categories of healthcare provider actors under the PHSA that generally lack information blocking disincentives under the current enforcement rule. While less categorical in nature, the current application of information blocking disincentives to only those healthcare providers participating in certain Medicare quality programs leaves many enforcement gaps. Applying information blocking disincentives more

uniformly, particularly to provider-types that are largely categorically excluded, such as laboratories and imaging centers, would incentivize all actors to leverage national exchange networks/frameworks in support of seamless data exchange, thereby promoting interoperability and enhancing patient access.

PA-2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?

CMS can encourage payers by mandating FHIR-based Payer-to-Payer and Provider Access APIs, offering incentives for early adoption, and aligning with TEFCA and other national health data sharing networks' API standards (based on HL7 DaVinci). Technical assistance and developer sandboxes, like those the eHealth Exchange provides, would accelerate implementation. These steps would enhance data sharing, leveraging national investment in TEFCA and the eHealth Exchange's connectivity to payers and providers.

The CMS should deploy an accurate, complete, and up-to-date national directory of health providers, payers, and other actors. Lack of a national directory is significantly increasing the burden of deploying API-based applications simply due to the difficulty in finding each relevant API. See *Broad Themes, topic 1* in this document for more detailed information on how a lack of national directory is chilling API usage.

Trust is a prerequisite for exchange. Providers are hesitant to share their patients' data if they know those data are being resold. eHealth Exchange does not store or resell patient data which we believe is important for the establishment of trust.

Finally, providers can, today, use a single connection for multiple purposes with Networked FHIR, as is supported now by the eHealth Exchange. See *Networked FHIR* for more information. This reduces providers' IT costs by reducing the number of technical connections they must maintain and secure, which will accelerate adoption of APIs.

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

We want to recognize that the CMS has mandated that TEFCA data holders MUST respond to requests for patient information. This is an important step as it allows mandatory responses and should act to stimulate developer interest in creating new digital health products. The CMS should continue to require responses for Individual Access Services TEFCA requests.

CMS should also provide for a national directory of healthcare actors See *Broad Themes, topic 1* in this document.

Finally, CMS should encourage TEFCA and other networks to use Networked FHIR since that will multiple developer value when creating API-based applications for patients and caregivers. See *Networked FHIR* in this document. Networked FHIR and a national directory are critically to this objective. Without these, API-based application developers must incur significantly higher expenses to identify APIs, and to connect to multiple API endpoints.

TD-2. a. What additional data would be most valuable if made available through CMS APIs?

An accurate and consistent national directory of providers, public and private payers, plans, organizations, and relationships would be most valuable for developers, enabling apps for care navigation and cost transparency. Currently the lack of such a national directory data is an almost insurmountable barrier for innovative solutions to be developed and brought to the market. See *Broad Themes, topic 1* in this document.

TD-2. c. What obstacles prevent accessing these data sources today?

The current CMS mandated FHIR Provider and Plan Directory API end points are not sufficient. Directory end points have differences in FHIR paging techniques, data models, inconsistent support of last-updated dates, different fundamental ways of representing provider to organization relationships, lack of tech support by the publishing organization, rate limiting resulting in weeks of time needed to obtain the directory data, disallowed login barriers, and more. Also, the CMS should mandate that ALL private and public payers and providers publish these directories. Notably, some entire states do not publish their directories using the CMS directories resulting in state-specific one-off solutions to obtain directory information. This likely would reduce costs to all health data sharing networks by standardizing these important data and reducing manual one-off integration work. We can share much more detailed information on the current interoperability barriers to obtaining directory information from the existing CMS mandated provider directory API end points. Finally, the CMS should fund work at IHE USA or HL7 for the creation of a more constrained FHIR Implementation Guide to eliminate the variations between APIs so that true directory interoperability is achieved. See *Broad Themes, topic 1* in this document.

TD-2. d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

The CMS should publish an accurate “directory of directories” listing each provider / payer / plan directory endpoint. Today, developers are forced to use manual means to discover these end points. This manual discovery process is expensive, time consuming, and this reduces developer interest due to the resources needed to find each FHIR directory end point to interface with. See *Broad Themes, topic 1* in this document.

TD-3. c. What impact would mandatory use of the OpenID Connect identity protocol have?

OIDC is not sufficient by itself; it has gaps that will likely reduce healthcare data sharing. The CMS should mandate that organizations implement HL7 FAST-recommended authentication instead, once complete, as it would allow for dynamic registration, sharing of mandatory healthcare-specified information such as the purpose of the request, and support intermediaries. OIDC is already in place today and it is acting as a barrier because the process

on onboarding to each data holder is a manual process. The FAST SSRAA alternative approach can automate this approach. The SSRAA standard is not finished at this time and needs production piloting and refinement to allow for FHIR security to scale nationally. The CMS should fund HL7 FAST SSRAA standards completion and production piloting to create a proven national scalable effort. The key capabilities will be a) Dynamic Client Registration, b) support for intermediaries, and c) sharing the context of the request (such as the purpose of use, the provider's human name, the organization making the request, etc.). The eHealth Exchange has deployed several FHIR production deployments, and we had to work around OIDC limits. See the *Mini Case Studies*.

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

The CMS can mandate that FHIR R4 (and above) be used as a standard for all data interoperability within the nation. Furthermore, the CMS can provide a roadmap to phase out older standards (see response TD-7 below) and to phase out proprietary and older standards-based APIs.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

The CMS should publish a nationwide provider directory adhering to the FHIR directory structures (Organization, Practitioner, PractitionerRole, etc.). However, directories are not useful unless they are kept current. The CMS should link significant financial incentives to payers, providers, and others to keep their directory entries accurate. Or the CMS should publish a national "directory of directories" to the various FHIR endpoints published by payers, providers, and plans. See *Broad Themes* topic #1, above, for more details about the obstacles this presents to developers.

TD-6. What unique interoperability functions does TEFCA perform?

While TEFCA has gone through the process of developing the policy and technical framework for trusted national exchange, and it has mandated responses for patient requests (Individual Access Services) it has missed an opportunity by not mandating Public Health, Research, and Insurance as mandatory-response use cases. TEFCA should also standardize Networked FHIR instead of point-to-point FHIR exchange. See *Networked FHIR* for more information on this topic.

TD-6. a. What existing alternatives should be considered?

eHealth Exchange, a non-profit founded for the public good as a 501(c)(3) by the CMS, is the oldest and largest health data sharing network in the country, to our knowledge. eHealth Exchange does not store patient data in a clinical data repository, and it does not resell patient

clinical data. eHealth Exchange is likely the only next generation clinical network live with Cambia in production with BCBSA and with Elevance in the process of implementing. **eHealth Exchange has never accepted a government or private grant and is financially sustainable**, and it presents a viable national network that continues to connect to all other available regional, vendor-lead, specialty, and national networks.

Since the TEFCA framework is not funded by federal government grants and has no publicized sustainability model, some question TEFCA's future. Self-funded eHealth Exchange has demonstrated value and sustainability, vendor neutrality, and a history of innovating while freely sharing its success and industry knowledge via IHE, HL7, and additional forums. eHealth Exchange has also demonstrated a consistent willingness to embrace and connect with all other networks for the better good. eHealth Exchange serves as a compelling example for national-scale connectivity for the nation just as it did in 2006 as an HHS initiative.

TD-6. b. Are there redundant standards, protocols or channels or both that should be consolidated?

Currently, there are at least EIGHT competing standards for healthcare and payment data exchange. These should be leveraged for a reasonable amount of time while each of these various domains is incentivized to move to FHIR transactions.

Standards such as NEMSIS for ambulance data exchange, NIEM for the Department of Justice, X12 for insurance, HL7 v2 for legacy labs and hospitals and public health reporting, IHE SOAP/SAML/XML for many national networks, DirectTrust secure email, HL7 v3 for national networks, and now FHIR. Additionally, some vendor-driven national networks use proprietary APIs. It is likely in the national best interest for the CMS to set a vision, and provide financial incentives, for a single standard, such as HL7 FHIR r4 (and above) with IHE FHIR standards for all these use cases. This ideally is a deliberate, nonpartisan, long-term project with a broadly approved timeline that leverages the interoperability deployed today, while giving us all a clear target for the future.

TD-7. d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

eHealth Exchange prefers highly structured FHIR-based USCDI for scalability, accuracy, and deterministic behaviors. As our billions of annual transactions demonstrate, structured data which include narrative text scales well. Less structured formats, while leveraging AI, will result in inconsistencies and processing burdens. CMS should prioritize structured data, with AI as a complement, ensuring alignment with our nation's standards. <u>Additionally, AI technologies are able to directly consume, produce, and understand JSON, ICD, CPT, LOINC, UMLS, SNOMED-CT and other structured data formats and value sets.</u> <u>Hence it would represent an unnecessary step backward to discard structured data in favor of unstructured data.</u> Ideally clinical data are both represented in structured **and** narrative formats. The CMS should encourage the adoption of standards and operational practices that mandate both structured and unstructured formats, as both serve important purposes.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

This capability has existed since 2006, in production, by the eHealth Exchange's open-standards based APIs.

Full EHI access via APIs, as eHealth Exchange has long supported in production, enable apps for predictive analytics, personalized care plans, and in the future these APIs could provide real-time SDOH integration and insurance data. Providers reduce errors, avoid expensive manual data entry, and patients gain holistic health views. CMS should mandate comprehensive API responses, leveraging our network's capabilities to drive innovation.

We also want to commend CMS and TEFCA for mandating responses to patient data requests (Individual Access Services). This is a national first and is critical as it requires all data holders in TEFCA to provide the patients' own data back to them for their benefit.

TD-13. a. What are the primary obstacles to this?

eHealth Exchange believes three of the obstacles are:

1. **Trust.** Organizations that re-sell patient clinical data undermine trust in sharing the entirety of patients' records using APIs. Since eHealth Exchange does not resell patient data, we can perhaps serve as one example of how patient trust can be established.
2. **Costs.** A secondary obstacle is the lack of CMS support for Networked FHIR. See *Networked FHIR.* Without Networked FHIR, organizations must incur high costs to establish and maintain multiple connections. With Networked FHIR, there is only a single connection to the rest of the nation for multiple purposes.
3. **Pushed content.** EHRs and most HIEs, do not allow "pushing" of medical data into a patient's record on inbox. See *Broad Themes, Topic #7.*

TD-13. b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

USCDI ensures standardized exchange but limits scope, while full EHI offers comprehensive data at the cost of complexity. eHealth Exchange's FHIR and SOAP APIs handle both, allowing the receiving system to choose which elements to process. CMS should expand USCDI gradually, leveraging networks to balance coverage and scalability. And this expansion should be driven by national priority use cases such as virtual clinical trials, adverse event follow-up, payer efficiency, and disease outbreak tracking.

TD-14. a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

The eHealth Exchange is connected to approximately 350 IHE SOAP connections and dozens of production FHIR end points. Most of these connections service multiple hospitals, clinics and providers. Our footprint is national with multiple connections in each state.

The data sources and providers are searchable via NPIs, organizational name, name aliases, various organizational identifiers, by state, hospital referral region, and all other FHIR-supported search criteria.

The endpoint types are IHE XCA, and HL7 FHIR.

TD-14. b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

HL7 FHIR r4 (and above), IHE SOAP Profiles (XCA, XUA, ATNA, CT) and IHE FHIR Profiles (MHD). They are secured by the latest versions of TLS using mutual authentication and X.509 certificates. IHE XCA Messages are further secured with message level digital signatures. All transactions are audit logged. But no central data repository of patient clinical data is stored reducing cybersecurity targets. We maintain a directory of electronic end points and participants to facilitate finding exchange partners.

eHealth Exchange supports Networked FHIR, which allows a single FHIR endpoint to be reused for multiple use cases, and for connections to the rest of the nation that is also available through single-connection Networked FHIR.

TD-14. c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

We interconnect with 58 regional and state HIEs, and approximately 50 other national networks including Carequality, CommonWell, and TEFCA. Our primary legal agreement is the eHealth Exchange DURSA. We also support interconnections via the TEFCA Common Agreement, and the Carequality Connected Agreement. eHealth Exchange's long-standing practice is to collaborate and connect with any valid health data or payment data sharing networks.

TD-15. a. How would increased use of bulk FHIR improve use cases and data flow?

Increased Bulk FHIR use, currently supported by eHealth Exchange, would enhance population health analytics, quality reporting measurement capture, and public health surveillance. It enables efficient data extraction, reducing provider and public health burden. eHealth Exchange has deployed Bulk FHIR for FDA public health use, and HEDIS measurement capture. Both of the use cases actually demonstrate the current capabilities of Bulk FHIR. CMS could mandate adoption of Bulk FHIR to improve data availability. The list of

bulk data exports, and their unique query identifiers, could be added to a national directory to facilitate interoperability and automated onboarding. Please also see the *Federal Agency Use of eHealth Exchange and FHIR for Population Health Bulk Queries, Adverse Reactions Notifications, and Case Follow-up* and *Use of the eHealth Exchange and Bulk FHIR for Digital HEDIS Measurement* Mini Case Studies in this document for two examples of Bulk FHIR usage.

TD-16. a. Do current rules encourage scalable network participation?

One significant barrier to scalable national FHIR interoperability is that lack of "system level" trust models. Most, if not all, EHRs require point-to-point manual authorization for read-only FHIR data access to their EHR. The CMS could help overcome this by mandating that EHRs provide system level trust support to enable automatic trust by any authorized member of TEFCA, eHealth Exchange, or other network participants. Unless the CMS requires adoption of system level FHIR trust for network participants, then national scalability for FHIR exchange is essentially not possible. TEFCA, Carequality, and eHealth Exchange all have implemented system-level trust, at scale, for IHE XCA transactions, for multiple years demonstrating its production viability. FHIR still requires manual onboarding for each connection. See the section on *Networked FHIR* for more information.

The lack of an authoritative national directory means each network has to maintain their own redundant list of participants, identifiers, and connectivity endpoints. See *Broad Themes, topic #1*.

Finally, the lack of formal support of intermediaries limits growth because each FHIR endpoint has to manually identified and manually onboarded with each other end-point. The eHealth Exchange was specifically established to eliminate this bottleneck. And, unfortunately, FHIR re-introduces this problem.

The CMS should mandate standards-based support for FHIR security, directories, and Networked FHIR to resolve these issues, otherwise FHIR at scale will continue to be hampered.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

As the prior rounds of funding demonstrate, it is not enough to fund only the development of initial health data sharing infrastructure. Many organizations received funding (eHealth Exchange did not) and many ceased to exist once the startup funding was exhausted. This created a "bubble" of non-viable HIEs. Instead, the CMS should offer long-term funding that increases the revenue streams of viable regional and national HIEs so that they can invest in better interoperability, cybersecurity, and more advanced features. This will take proven HIEs and allow them to move from a mode of minimal financial survivability to a mode of thriving and adding increasing value to the national infrastructure. The CMS should also offer innovation grants to provide for DARPA-like high-risk high rewards next-generation interoperability.

TD-18. b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?

ONC and CMS could audit provider compliance and could publish violation reports to deter information blocking.

TD-18. c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

Laboratories and imaging centers are two categories of healthcare provider actors under the PHSA that generally lack information blocking disincentives under the current enforcement rule. While less categorical in nature, the current application of information blocking disincentives to only those healthcare providers participating in certain Medicare quality programs leaves many enforcement gaps. Applying information blocking disincentives more uniformly (particularly to provider-types that are largely categorically excluded, such as laboratories and imaging centers) would incentivize all actors to leverage national exchange networks/frameworks in support of seamless data exchange, thereby promoting interoperability and enhancing patient access.

TD-19. Regarding price transparency implementation:

TD-19. a. What are current shortcomings in content, format, delivery, and timeliness?

Price transparency data are not shared in FHIR format, does not offer real-time updates, and patient-friendly delivery, hindering use of those data. Content often omits negotiated rates, and delivery via the current 'machine readable' format is inaccessible. CMS should mandate FHIR-based price APIs to ensure timeliness and usability. The pricing transparency APIs should then be listed in the national provider director, see *Broad Themes, topic #1*. These two changes (FHIR API access and directory listings) would likely result in dramatically more use and availability of pricing transparency information.

Workflows, such as care navigation, provider selection, and prior authorization, would benefit most. eHealth Exchange's APIs, and other national networks, could deliver price transparency data to patient apps. Patients could then compare providers, and payers could streamline approvals. CMS should enable and encourage integration of price data into our national networks.

TD-19. b. Which workflows would benefit most from functional price transparency?

FHIR-based price APIs, real-time updates, and patient-facing tools would be most valuable, enabling eHealth Exchange and TEFCA to support cost-conscious patient care applications. Providers would benefit from streamlined referrals, payers from efficient contracting, and CMS from reduced costs. Standardized, FHIR-based, formats would ensure scalability.

TD-19. c. What improvements would be most valuable for patients, providers, or payers, including CMS?

Adoption of FHIR for sharing of these data via APIs would have the largest impact.

1.VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

CMS should fund HL7 and IHE standards focused on the use of AIs in healthcare. Currently, there are insufficient standards to manage and convey the usage of AI in healthcare and claims processing. In order for AI actors to be useful in healthcare they must be trustworthy. No existing healthcare standard allows for the expression of the verifiable use of AI as an actor to a healthcare transaction such as a cryptographic checksum ensuring the AI model (the 'weights') and the AI execution environment are authentic, intact, and from an authorized source. Another example is lack of HL7 and IHE standards regarding audit logging of AI transactions (content creation, patient chats, summaries, clinical interpretations for physician review, etc.). These missing standards are a prerequisite for AI adoption in healthcare and they are a significant barrier for AI adoption as the lack of AI-centric standards prevent AIs from being curated as approved actors in the ecosystem.

1.VB-3. What are essential health IT capabilities for value-based care arrangements?

Essential capabilities include care planning, adherence to best practices (esp. for chronic conditions), patient event notifications, and quality measurement, all technical capabilities supported by eHealth Exchange's FHIR APIs. Our network enables real-time data extraction and attribution, critical for ACOs. Remote device interoperability and patient empowerment tools, integrated via our APIs, could ensure comprehensive care, aligning with CMS's value-based care goals. Currently the eHealth Exchange, TEFCA, and other national networks remain underutilized for value-based care. CMS should incentivize the use of national networks (a largely untapped national resource) to increase quality and reduce the costs of value-based care.

Social Indicators (or Determinants) of Health are crucial for a large minority of patient care in the nation, but are not available electronically. May patients are treated optimally in care settings but then go back to community situations where the patient is simply unable to adhere to proper care guidelines due to food, financial, social, transportation, or housing concerns. Providers that can become aware of these indicators in a timely manner can intervene, provide appropriate referrals, administer FHIR-based social risk assessments, and share data with others, to the benefit of the patient. This would also have the added benefit of likely more efficient care, helping us all achieve the quadruple aim.

Conclusion

The country is now well positioned to capitalize on the massive investment the United States has made in improved clinical and payment interoperability over the last 10 to 15 years. The quadruple aim remains highly relevant today and it is within our reach. CMS can play a vital and unique role in achieving the quadruple aim by leveraging its payment power to create the proper incentives to align the national strategy around current best practices.

Key recommendations:

1. Create an accurate and complete national directory of all healthcare actors and technical endpoints and provide financial incentives for directory accuracy
2. Mandate payers and providers use national networks for electronic Prior Authorization and other payer transactions
3. Encourage adoption and use of FHIR for Public Health use cases
4. Mandate use of Bulk FHIR for Public Health and other use cases
5. Begin to defragment and incorporate Social Indicators / Determents of Health actors and data in national exchanges
6. Mandate the use of computable expressions of patient consent and authorization forms
7. Facilitate solutions to the remaining challenges preventing national-scale FHIR deployment
8. Set a national direction that gradually migrates from eight or more competing healthcare data and transaction standards to FHIR r4 and above
9. Leverage and support self-sustaining HIEs that do not profit from selling patient data
10. Use the early successes, and identified gaps, of the eHealth Exchange in deploying multiple production FHIR use cases for lessons to help clarify the national vision
11. Leverage national networks and regional HIEs for the electronic collection of HEDIS measurement data elements

More information is available on the items briefly covered in this RFI response; eHealth Exchange welcomes the opportunity to clarify or provide specific examples so CMS can learn from our experiences as the first national network to support FHIR in production. eHealth Exchange sincerely hopes the information we have shared herein assists CMS efforts.

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Prepared by Eric Heflin. Any errors or omissions are my own.

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